# Roy (Yuanxi) Cheng

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## Education

**University of California, Los Angeles (UCLA)** Sep 2023 – Mar 2027

*Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering* *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Machine Learning, Neural Networks & Deep Learning, NLP, Data Mining, Data Analysis & Regression, Data Structures & Algorithms, Probability & Statistical Inference, Databases, Monte Carlo, Optimization

## Work and Leadership Experience

**Capital One** McLean, Virginia

*Product Management and Analysis Intern* *Jun 2026 – Present*

- Leveraged SQL and Python to analyze workforce data, identifying factors tied to successful organizational transitions
- Built a FAISS semantic index over O*NET occupations with BGE embeddings for LLM retrieval of DWA candidates
- Designed a two-stage LLM pipeline with a review denoiser and DWA evidence matcher to produce ranked skills menus

**Data Science Union** Los Angeles, California

*External Vice President* *Dec 2025 – Present*

- Directed client partnerships, marketing, and professional development initiatives to expand opportunities across members
- Managed 3 directors and coordinated executive priorities, ensuring seamless execution of quarterly and annual objectives
- Cultivated industry and alumni partnerships to secure client projects, increase visibility, and expand career programming

**Kohl's** Los Angeles, California

*Project Manager and Data Analyst* *Jun 2025 – Mar 2026*

- Analyzed 8.5M+ logistics records using SQL and Python, resolving inconsistencies and validating supply chain data
- Conducted EDA in Vertex AI Workbench, engineering predictive features to enhance shipment transit time forecasting
- Developed and deployed two-stage XGBoost forecasting model, improving shipment arrival prediction MAE by 83%

**ACTUM Digital** Prague, Czech Republic

*Data Analyst and AI Intern* *Jun 2025 – Aug 2025*

- Managed project workflows in Jira/Confluence, presenting structured weekly updates that improved delivery timeline
- Engineered Python scripts with agentic methods to filter files, generate metadata, and embed into Milvus vector database
- Authored LLM prompting and integrated MCP server to create AI assistant, co-presenting demo to ACTUM executives

**Million Dollar Baby** Los Angeles, California

*Data Analysis Consultant* *Apr 2025 – Jun 2025*

- Analyzed $10M+ ad spend data with SQL to extract insights that informed growth strategy and investment decisions
- Presented CAC and RoAS assessments across attribution windows to guide stakeholders on marketing spend allocation
- Built a dashboard of 20+ marketing KPIs in Excel to highlight performance trends to brand managers and stakeholders

## Projects and Research

**Adrenaline and Cognitive Fatigue Effects on Athletic Performance** | *R, ggplot2, tidyverse, dplyr*

- Designed $2^3$ factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Verified model assumptions with log transformation and ggplot2 diagnostics, confirming normality and homoscedasticity

**Biomedical Tissue Imaging** | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article analyzing CNN-based segmentation models and AP metrics to evaluate cytoplasm boundary precision
- Engineered data pipeline to convert 4k+ CytoNuke annotations into COCO to allow training across 3 model architectures
- Improved cytoplasm segmentation by training YOLOv8 in PyTorch, increasing AP50 by 5% over Cyto-RCNN

**COVID Pandemic Effects on Secondary Education** | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records by merging multi-level datasets, enabling accurate visualization and analysis in Tableau
- Built time series visualizations by aggregating 4 years of test scores, revealing 15% avg decline in student performance
- Created interactive Tableau map for research website by linking geographic data to score changes across 50 U.S. states

## Technical Skills

**Languages**: Python, SQL, R, C++, JavaScript, Java
**Libraries/Frameworks**: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, pandas, NumPy, Matplotlib, Seaborn, ggplot2
**Tools & Platforms**: Vertex AI, GCP, BigQuery, Snowflake, Tableau, Git, GitHub, Jupyter Notebooks